Pisidium amnicum
(greater European peaclam)
Mollusks-Bivalves
Exotic

Collection Info
Point Map
Species Profile
Animated Map

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Pisidium amnicum (Müller, 1774)

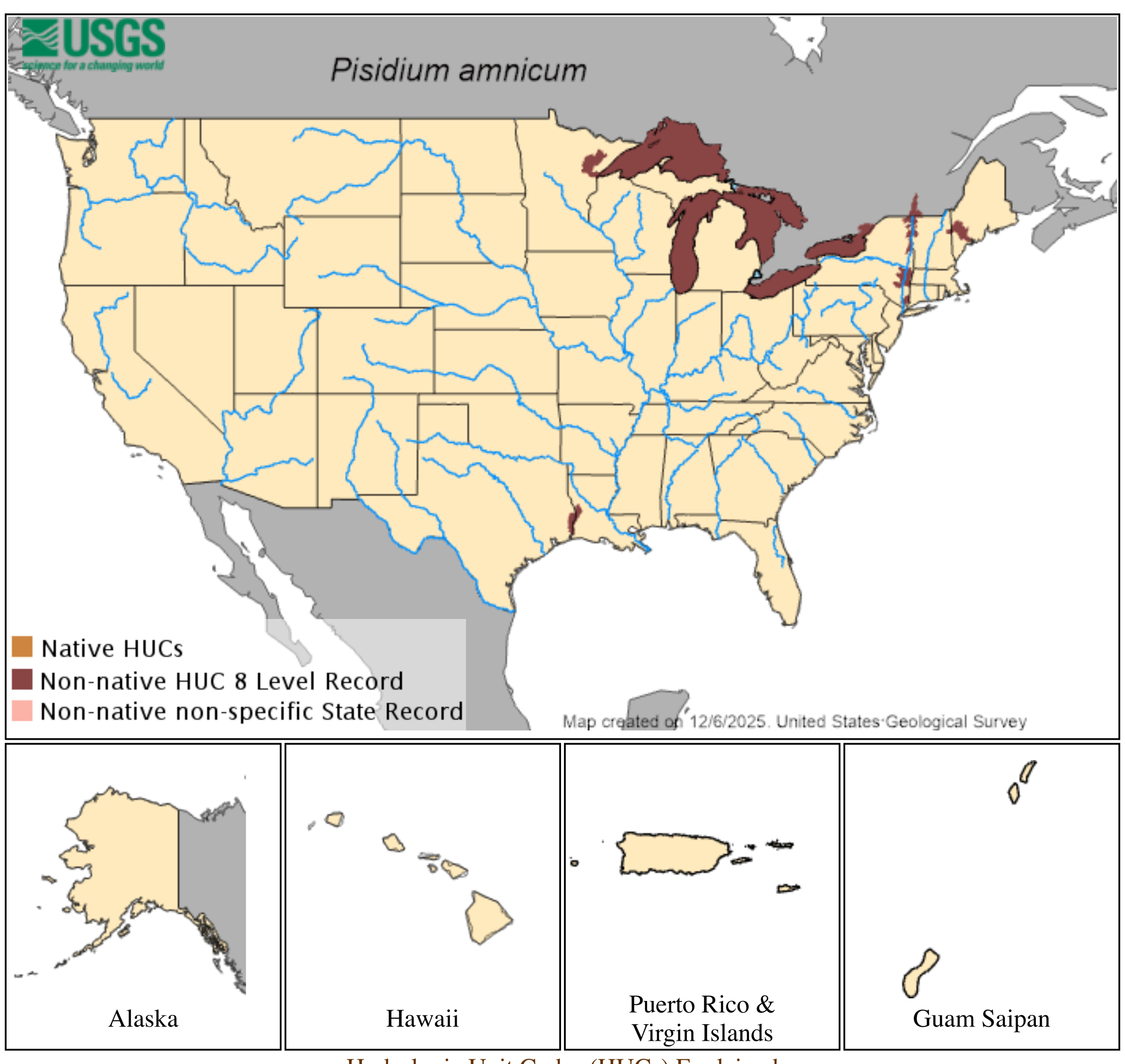
Common name: greater European peaclam

Taxonomy: available through ITIS

Identification: This small bivalve has a height to length ratio of 0.74–0.81 and is relatively long, oval shaped, and heavily striated with a shiny yellow or brown epidermis. The beaks are located towards the posterior by about 2/3 of the total shell length. Inside the shell, the cardinal teeth are closer to the anterior lateral teeth than to the posterior lateral teeth, the 2nd cardinal tooth is a thick peg covered by the thinner 4th cardinal, and the 3rd cardinal curves around the 2nd cardinal. In live specimens there is only an anal siphon (Clarke 1981, Herrington 1962, Mackie et al. 1980, Pennak 1989).

Size: varies from 8.8 to 11.9 mm in length (Herrington 1962, Holopainen 1979, Holopainen et al. 1997, Vincent et al. 1981).

Native Range: *Pisidium amnicum* is widely distributed in Eurasia and North Africa between Naples, Siberia, and Algiers (Mackie 2000, Por et al. 1986, Vincent et al. 1981).



Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Pisidium amnicum* are found here.

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
IL	2015	2015	1	Lake Michigan
LA	1962	1962	1	Lower Sabine
ME	1967	1967	1	Lower Androscoggin River
MI	1899	1899	1	Lake Huron
MN	1985	2016	2	Lake Superior; St. Louis
NJ	1962	1962	1	Mid-Atlantic Region
NY	1897	1996	5	Headwaters St. Lawrence River; Lake Champlain; Lake Ontario; Lower Hudson; Middle Hudson
PA	1962	1962	*	
VT	1962	1962	1	Richelieu River
WI	2016	2018	2	Lake Superior; St. Louis

Table last updated 12/8/2025

† Populations may not be currently present.

* HUCs are not listed for states where the observation(s) cannot be approximated to a HUC (e.g. state centroids or Canadian provinces).

Ecology: Found in freshwater lakes and slow-moving rivers with soft bottoms; water temperatures of 1–21ºC. *Pisidium amnicum* is typically a rheophilic species in its native range but can also occur in lakes. It prefers sand but has also been recorded on mud and gravel. It can survive anoxic conditions under ice cover but may be limited in some upper river reaches where temperatures do not exceed 15–17ºC in July. *Pisidium amnicum* is capable of closing its shell to induce anoxia, metabolic quiescence, and anaerobiosis, and can survive for 200 days at 0ºC. It occurs down to 30 m in Europe but only down to 10 m in the Great Lakes. Densities in Europe have reached around 1000–3300 clams per m^2.

In the St. Lawrence River, Canada, where it has been introduced, it is often found living in littoral zones in association with the introduced snail *Bithynia tentaculata* and the oligochaete *Sparganophilus tamesis* (Bishop and Hewitt 1976, Holopainen 1979, 1987, Mackie et al. 1980, Vincent et al. 1981, Dyduch-Falniowska 1982, Piechocki and Luczak 1989, Holopainen and Penttinen 1993, Grabow 1994, Zettler 1997, 1998, Holopainen et al. 1997, Mackie 2000).

Pisidium amnicum can live up to 3 years, mature at 4 mm (sexually mature as early as 3 months old in Europe). It is hermaphroditic, ovoviviparous, and can undergo cross-fertilization. Eggs incubated in a brood-sac in the parent; embryos develop and are released as miniature adults. In Europe it is often semelparous, reproducing once in a lifetime. In the St. Lawrence River it is iteroparous, reproducing twice, once at age 2 and once at age 3. Recruitment takes place when water temperatures reach 15–20ºC. Maturation of individuals and egg-laying occur between July and October, and eggs are brooded for around 9–10 months. The number of embryos per adult varies from 5–29. Lifespan is typically 1–3 years (Araujo and Ramos 1999, Araujo et al. 1999, Holopainen 1979, Holopainen et al. 1997, Vincent et al. 1981).
Pisidium amnicum larvae may be distributed by ruminants via excrement. Adult clams in particular can be hosts to digenean parasites in Eurasia, such as: *Bunodera lucipercae*, *Palacerochis crassus*, *Phyllodistomum elongatum*, and *Crepidostomum* sp. Parasites may castrate their hosts. Semelparity could be a result of castration (Holopainen et al. 1997, Rantanen et al.1998, Sturm 2000, Zhokov 1990).

Pill clams are filter feeders (suspension feeders on algae and bacteria), living in the sediments and obtaining nutrition from the substrate and the water column. This species especially favors diatoms (Holopainen 1979, Mackie 2000).

Means of Introduction: *Pisidium amnicum* was very likely introduced in solid ballast, which was used in the early 1900s in ships entering the Great Lakes (Mills et al. 1993; Grigorovich et al. 2003).

Status: Established where recorded, but at low densities in some regions. *Pisidium amnicum* could also occur in other inland waters within the Great Lakes basin (Clarke 1981, Grigorovich et al. 2003, Mackie et al. 1980, Mills et al. 1993).

Impact of Introduction: The impacts of this species are currently unknown, as no studies have been done to determine how it has affected ecosystems in the invaded range. The absence of data does not equate to lack of effects. It does, however, mean that research is required to evaluate effects before conclusions can be made.

Remarks: In some parts of its native range *P. amnicum* is considered endangered (Beran 1998, Sturm 2000).

References (click for full references)

Araujo, R., and M.A. Ramos. 1999. Histological description of the gonad, reproductive cycle, and fertilization of *Pisidium amnicum* (Muller, 1774) (Bivalvia: Sphaeriidae). Veliger 42(2):124-131.

Araujo, R., M.A. Ramos, and R. Molinet. 1999. Growth pattern and dynamics of a southern peripheral population of *Pisidium amnicum* (Muller, 1774) (Bivalvia: Sphaeriidae) in Spain. Malacologia 41(1):119-137.

Bishop, M.J., and S.J. Hewitt. 1976. Assemblages of *Pisidium* spp. (Bivalvia: Sphaeriidae) from localities in eastern Poland. Freshwater Biology 6(2):177-182.

Beran, L. 1998. Molluscs (Gastropoda; Bivalvia) of the wetlands in the Libechovka and Psovka brook basin. Casopis Narodniho Muzea Rada Prirodovedna 167(1-4):43-51.

Clarke, A.H. 1981. French translation by A. La Rocque. Les Mollusques d'eau douce du Canada. Musée national des sciences naturelles, Musées nationaux du Canada. 447 pp.

Dyduch-Falniowska, A. 1982. Oscillations in density and diversity of *Pisidium* communities in 2 biotopes in southern Poland. Hydrobiological Bulletin 16(2-3):123-132.

Grabow, K. 1994. Mollusca of Salzgitter branch canal (Mittellandkanal) near Brunswick, Lower Saxony. Braunschweiger Naturkundliche Schriften 4(3):485-496.

Grigorovich, I.A., A.V. Korniushin, D.K. Gray, I.C. Duggan, R.I. Colautti, and H.J. MacIsaac. 2003. Lake Superior: an invasion coldspot? Hydrobiologia 499(1-3):191-210.

Heard, W.H. 1962. The Sphaeriidae (Mollusca: Pelecypoda) of the North American Great Lakes. American Midland Naturalist 67(1):194-198.

Herrington, H.B. 1962. A revision of the Sphaeriidae of North America (Mollusca: Pelecypoda). Museum of Zoology, University of Michigan, Miscellaneous Publications 118. 74 pp 7 plates.

Holopainen, I.J. 1979. Population dynamics and production of *Pisidium* species (Bivalvia, Sphaeriidae) in the oligotrophic and mesohumic lake Pääjärvi, southern Finland. Archiv für Hydrobiologie 54(4): 466-508.

Holopainen, I.J. 1987. Seasonal variation of survival time in anoxic water and the glycogen content of *Sphaerium corneum* and *Pisidium amnicum* (Bivalvia: Pisididae). American Malacological Bulletin 5(1):41-48.

Holopainen, I.J., and O. Penttinen. 1993. Normoxic and anoxic heat output of freshwater bivalves *Pisidium* and *Sphaerium*. Oecologia 93: 215-223.

Holopainen, I.J., S. Lamberg, E.T. Valtonen, and J. Rantanen. 1997. Effects of parasites on life history of the freshwater bivalve, *Pisidium amnicum*, in eastern Finland. Archiv fur Hydrobiologie 139(4):461-477.

Mackie, G.L. 2000. Ballast water introductions of Mollusca in R. Claudi, and J.H. Leach, eds. Nonindigenous Freshwater Organisms: Vectors, Biology and Impacts. CRC Press LLC, Boca Raton, Florida 219-254.

Mackie, G.L., D.S. White, and T.W. Zdeba. 1980. A guide to freshwater mollusks of the Laurentian Great Lakes with special emphasis on the genus *Pisidium*. Environmental Research Laboratory, Office of Research and Development, U. S. Environmental Protection Agency, Duluth, Minnesota 55804. 144 pp.

Mills, E.L., J.H. Leach, J.T. Carlton, and C.L. Secor. 1993. Exotic species in the Great Lakes: a history of biotic crises and anthropogenic introductions. Journal of Great Lakes Research 19(1):1-54.

Pennak, R. 1989. Fresh-water Invertebrates of the Unites States, 3rd ed. Protozoa to Mollusca. John Wiley & Sons, Inc., New York, New York State. 628 pp.

Piechocki, A., and C. Luczak. 1989. Sphaeriidae (Bivalvia: Eulamellibranchia) of the upper and middle course of the Wieprz River, southeast Poland. Przeglad Zoologiczny 33(4):59-566.

Por, F.D., H.J. Bromley, Ch. Dimentman, G.N. Herbst, and R. Ortal. 1986. River Dan, headwater of the Jordan, an aquatic oasis of the Middle East. Hydrobiologia 134:121-140.

Rantanen, J.T., E.T. Valtonen, and I.J. Holopainen. 1998. Digenean parasites of the bivalve mollusk *Pisidium amnicum* in a small river in eastern Finland. Diseases of Aquatic Organisms 33(3):201-208.

Sturm, R. 2000. The fresh water molluscs in waters and small swamps of the Postalm area (Tennengau, Salzburg). Linzer Biologische Beitraege 32(2):1235-1246.

Trebitz, A.S., C.W. West, J.C. Hoffman, J.R. Kelly, G.S. Peterson, and I.A. Grigorovich. 2010. Status of non-indigenous benthic invertebrates in the Duluth-Superior Harbor and the role of sampling methods in their detection. Journal of Great Lakes Research 36: 747-756.

US Army Corp of Engineers (USACE). 2012. Molluscicides. Accessed 10/28/13. http://glmris.anl.gov/documents/docs/anscontrol/Molluscicides.pdf

Vincent, B., G. Vaillancourt, and N. Lafontaine. 1981. Cycle of development, growth, and production of *Pisidium amnicum* (Mollusca: Bivalvia) in the St. Lawrence River (Quebec). Canadian Journal of Zoology 59(12):2350-2359.

Zettler, M.L. 1997. The aquatic malacofauna (Gastropoda et Bivalvia) in the catchment area of a North German Lowland River, the Warnow. Limnologica 26(3):327-337.

Zettler, M.L. 1998. The aquatic molluscs in the drainage area of the River Peene (North-East Germany). Malakologische Abhandlugen (Dresden) 19(1):127-138.

Zhokov, A.E. 1990. Seasonal and age-related characteristics of trematode parthenite invasion of a bivalve mollusk *Pisidium amnicum* population. Byulleten' Moskovskogo Obshchestva Ispytatelei Prirody Otdel Biolgicheskii 95(6):43-52.

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